

CANDIDATE
NAME

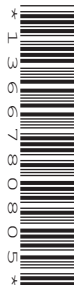
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CENTRE
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9231/33

October/November 2022

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

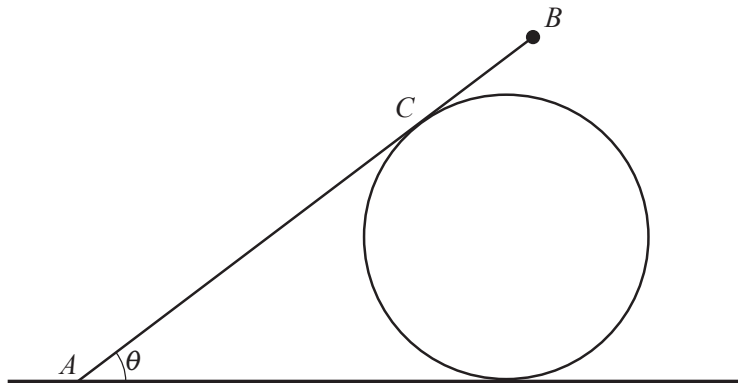
DC (SLM) 316816
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[Turn over

This image shows a full page of blank handwriting practice paper. It features approximately 20 horizontal blue lines spaced evenly down the page. Each line is composed of small dots, giving it a dashed appearance. The lines are designed to help children learn letter height and placement while writing. There are no margins, text, or other markings on the page.

- [illegible]

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- This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.



A particle of weight kW is attached to the rod at B . The rod is about to slip. The normal reaction between the rod and the cylinder is N .

- (a) Show that $N = \frac{8}{15} W(1 + 2k)$. [2]

[illegible]

[Turn over

- 4 A particle of mass 0.5 kg moves along a horizontal straight line. Its velocity is $v\text{ ms}^{-1}$ at time $t\text{ s}$. The forces acting on the particle are a driving force of magnitude 50 N and a resistance of magnitude $2v^2\text{ N}$. The initial velocity of the particle is 3 ms^{-1} .

(a) Find an expression for v in terms of t .

[7]

[illegible]

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(b) Deduce the limiting value of v . [1]

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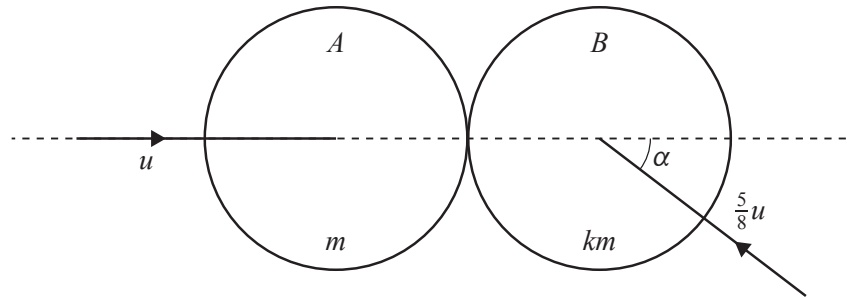
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- [illegible]

At Q the particle P becomes detached from the string.

- (b)** In the subsequent motion, find the greatest height reached by P above the level of the lowest point of the circle. [4]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.



Two uniform smooth spheres A and B of equal radii have masses m and km respectively. The two spheres are moving on a horizontal surface with speeds u and $\frac{5}{8}u$ respectively. Immediately before the spheres collide, A is travelling along the line of centres, and B 's direction of motion makes an angle α with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{2}{3}$ and $\tan \alpha = \frac{3}{4}$.

After the collision, the direction of motion of B is perpendicular to the line of centres.

- (a) Find the value of k . [4]

[illegible]

- [illegible]

- 7 A particle P is projected with speed $V \text{ m s}^{-1}$ at an angle 75° above the horizontal from a point O on a horizontal plane. It then moves freely under gravity.

(a) Show that the total time of flight, in seconds, is $\frac{2V}{g} \sin 75^\circ$. [2]

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A smooth vertical barrier is now inserted with its lower end on the plane at a distance 15 m from O . The particle is projected as before but now strikes the barrier, rebounds and returns to O . The coefficient of restitution between the barrier and the particle is $\frac{3}{5}$.

(b) Explain why the total time of flight is unchanged. [1]

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- (c) Find an expression for V in terms of g . [7]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

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